

Question 1: For the reaction $A \rightarrow \text{Products}$, the rate constant is $5.0 \times 10^{-2} \text{ Ms}^{-1}$. The reaction follows the integrated rate law:

- a. $\ln\left(\frac{[A]_t}{[A]_0}\right) = -kt + \ln[A]_0$
- b. $[A]_t = -kt + [A]_0$
- c. $\ln\left(\frac{[A]_t}{[A]_0}\right) = kt$
- d. $\frac{1}{[A]_t} = kt + \frac{1}{[A]_0}$
- e. $\frac{1}{[A]_t} = kt + \frac{1}{[A]_0}$

Question 2: Which of the following best describes all the intermolecular forces exhibited by a pure sample of $\text{HOCH}_2\text{CH}_2\text{OH}$?

- a. Dipole-dipole and hydrogen bonding
- b. Dispersion and dipole-dipole
- c. Dispersion only
- d. Dispersion, dipole-dipole, and hydrogen bonding
- e. Dispersion and hydrogen bonding

Question 3: Consider the following pairs of liquids. Which pairs are **miscible**?

1. Benzene (C_6H_6) and hexane (C_6H_{12})
 2. Water and methanol (CH_3OH)
 3. Water and hexane
- a. 2 only
 - b. 1 only
 - c. 1, 2 only
 - d. 1, 2, 3
 - e. 2, 3 only

Question 4: Which one of the following liquids would be expected to have the highest viscosity at room temperature?

- a. C_6H_6
- b. CH_2Cl_2
- c. $\text{HOCH}_2\text{CH}_2\text{CH}_2\text{OH}$
- d. $\text{HOCH}_2\text{CH}_2\text{CH}_2\text{CH}_3$

Question 5: A substance at a temperature greater than its critical temperature is called:

- a. Solid
- b. Liquid
- c. Vapor
- d. Supercritical liquid

Question 6: Rank the following compounds from weakest intermolecular forces to strongest intermolecular forces:

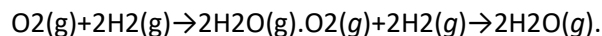
1. H_2O
2. N_2
3. H_2S

- **Answer:** $\text{N}_2 < \text{H}_2\text{S} < \text{H}_2\text{O}$

Question 7: Which of the following phase changes is(are) endothermic?

1. Melting
 2. Vaporization
 3. Sublimation
 4. Condensation
 5. Deposition
 6. Freezing
- a. 4 and 6 only
 - b. 1 and 2 only
 - c. 4, 5, and 6
 - d. 1, 2, and 3

Question 8: The following graph shows the kinetics curves for the reaction of oxygen with hydrogen to form water:



Which curve represents hydrogen ($\text{H}_2(\text{g})$)?

- a. The black curve
- b. The gray curve
- c. Either the gray or the black curve
- d. Any of these curves could be hydrogen

- e. The dashed curve

Question 9: A 250 mL solution containing 10.0 g of a polymer in toluene had an osmotic pressure of 0.055 atm at 27°C. What is the molar mass of the polymer?

- a. 18000 g/mol
- b. 38000 g/mol
- c. 15000 g/mol
- d. 26000 g/mol
- e. 32000 g/mol

Question 10: What types of intermolecular forces exist between NH_3 and H_2O ?

- a. Dispersion forces, hydrogen bonds, and ion-dipole forces
- b. Dispersion forces
- c. Dispersion forces and ion-dipole forces
- d. Dispersion forces and hydrogen bonds
- e. Dispersion forces, dipole-dipole forces, and hydrogen bonds

Question 11: Which probably has the lowest boiling point at 1.00 atm?

- a. H_2Te
- b. H_2S
- c. H_2Se
- d. H_2O

Question 12: Given the following data, determine the rate law for the reaction $\text{X} + \text{Y} \rightarrow \text{Z}$:

Exp [X] (M) [Y] (M) Rate (M/s)

1	0.0300	0.0100	3.4×10^{-4}
2	0.0150	0.0100	8.5×10^{-5}
3	0.0150	0.0400	3.4×10^{-4}

- a. $\text{Rate} = k[\text{X}]^2[\text{Y}]$
- b. $\text{Rate} = k[\text{X}][\text{Y}]$
- c. $\text{Rate} = k[\text{X}]^2[\text{Y}]^2$
- d. $\text{Rate} = k[\text{X}][\text{Y}]^2$

Question 13: Which of the following liquids will have the lowest freezing point?

- a. Aqueous CoI_2 (0.030 m)
- b. Pure H_2O
- c. Aqueous CaH_2O_6 (0.60 m)
- d. Aqueous FeI_3 (0.24 m)
- e. Aqueous KFKF (0.50 m)

Question 14: Which of the following choices has the compounds correctly arranged in order of increasing solubility in water? (least soluble to most soluble)

- a. $\text{CCl}_4 < \text{CHCl}_3 < \text{CH}_3\text{OH} < \text{CHCl}_3 < \text{CH}_3\text{OH}$
- b. $\text{CHCl}_3 < \text{CCl}_4 < \text{NaNO}_3 < \text{CHCl}_3 < \text{CCl}_4 < \text{NaNO}_3$
- c. $\text{CH}_3\text{OH} < \text{CH}_4 < \text{LiF} < \text{CH}_3\text{OH} < \text{CH}_4 < \text{LiF}$
- d. $\text{CH}_4 < \text{NaNO}_3 < \text{CHCl}_3 < \text{CH}_4 < \text{NaNO}_3 < \text{CHCl}_3$

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14
A	D	C	C	D	$\text{N}_2 < \text{H}_2\text{S} < \text{H}_2\text{O}$	D	E	B	E	B	D	D	A